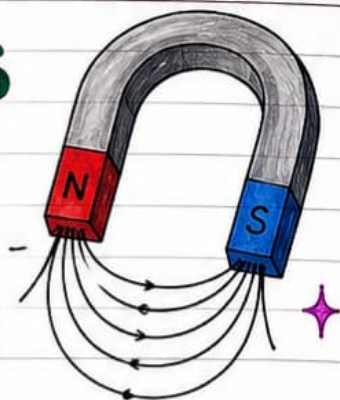


CHAPTER

4

Moving Charges and Magnetism

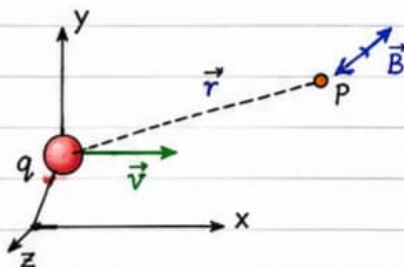
Magnetism arises due to motion of charges. A moving charge produces magnetic field which exerts a force on other moving charges.



① MAGNETIC FIELD DUE TO A MOVING CHARGE

When a charge q moves with velocity $\vec{v}$, it produces a magnetic field $\vec{B}$ at a point in space.

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q(\vec{v} \times \hat{r})}{r^2}$$



$\vec{v}$ = velocity of charge

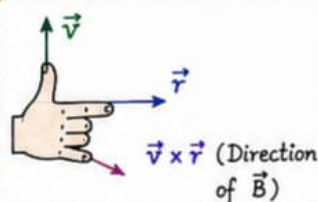
$\vec{r}$ = position vector

$r = |\vec{r}|$

μ_0 = permeability of free space

Direction of $\vec{B}$:

Given by right hand rule for $\vec{v} \times \hat{r}$



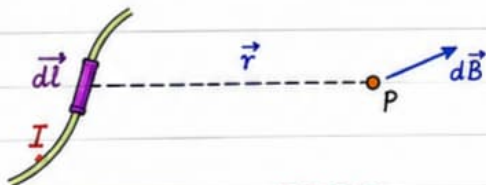
Key Points

- ★ If charge is at rest ($\vec{v} = 0$), $\vec{B} = 0$
- ★ $\vec{B} \propto q$, $\vec{B} \propto v$, $\vec{B} \propto \sin \theta$
- ★ Maximum $\vec{B}$ when $\vec{v} \perp \vec{r}$
- ★ Minimum $\vec{B}$ when $\vec{v} \parallel \vec{r}$

② MAGNETIC FIELD DUE TO A CURRENT ELEMENT

Consider a small current element $I d\vec{l}$. The magnetic field $\vec{B}$ at point P is given by Biot-Savart Law.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$



I = current

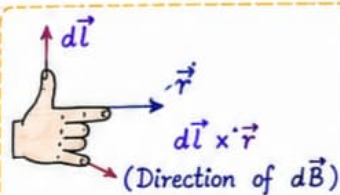
$d\vec{l}$ = length element

$\vec{r}$ = position vector

$r = |\vec{r}|$

Direction:

Given by right hand rule for $d\vec{l} \times \vec{r}$



Important

Total magnetic field due to a current carrying conductor is obtained by vector sum (integration) of all $d\vec{B}$.

Units

$$\text{Tesla (T)} = \frac{\text{Weber}}{\text{m}^2} = \frac{\text{N}}{\text{A} \cdot \text{m}}$$

Remember

Moving charge $\rightarrow$ Magnetic field

Current $\rightarrow$ Magnetic field

Changing magnetic field $\rightarrow$ Electric field (Ch-6)

Applications

- Basis of electric motors
- Loudspeakers
- Magnetic storage devices
- MRIs in medical imaging



Quick Revision

- ① Moving charge produces $\vec{B}$.
- ② Direction by right hand rule.
- ③ Magnitude $\propto \frac{qv \sin \theta}{r^2}$
- ④ Current element field by Biot-Savart Law.



③ BIOT-SAVART LAW

Biot-Savart law gives the magnetic field at a point due to a current element.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

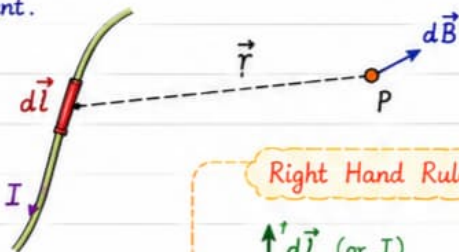
I = current

$d\vec{l}$ = current element vector

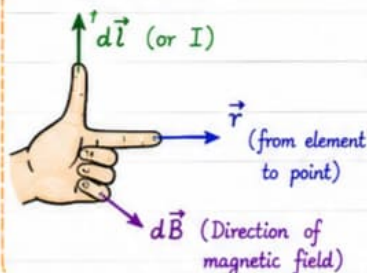
$\vec{r}$ = position vector from $d\vec{l}$ to point P

$r = |\vec{r}|$

μ_0 = permeability of free space



Right Hand Rule

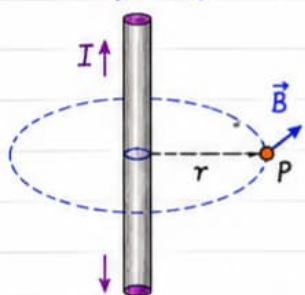


Key Points

- ★ $d\vec{B}$ is perpendicular to both $d\vec{l}$ and $\vec{r}$.
- ★ Magnitude depends on I , $d\vec{l}$, $\sin\theta$ and r^2 .
- ★ We have to integrate $d\vec{B}$ over the whole conductor to get net $\vec{B}$.

④ MAGNETIC FIELD DUE TO A LONG STRAIGHT CURRENT CARRYING CONDUCTOR

For a long straight wire carrying current I .



$$B = \frac{\mu_0 I}{2\pi r}$$

(direction given by right hand thumb rule)

Direction:

If thumb shows direction of current, curled fingers show the direction of $\vec{B}$ around the wire.

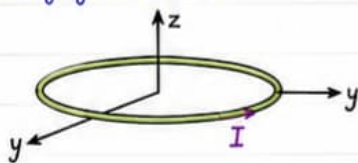


Important

- $B \propto I$ (directly)
- $B \propto \frac{1}{r}$ (inversely)
- Field lines are concentric circles around the wire.

⑤ MAGNETIC FIELD DUE TO A CIRCULAR CURRENT LOOP

For a circular loop of radius R carrying current I .



At the centre:

$$B = \frac{\mu_0 I}{2R}$$

(along the axis of the loop)

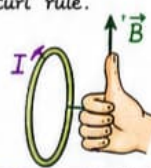
On the axis of the loop (at a distance x):

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

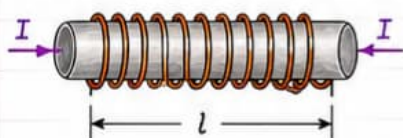
(along the axis)

Direction:

Use right hand curl rule.



⑥ MAGNETIC FIELD DUE TO A SOLENOID



Inside a long solenoid:

$$B = \mu_0 n I$$

(uniform and along the axis)

n = number of turns per unit length ($n = \frac{N}{l}$)

N = total number of turns

l = length of solenoid

Key Points

- Field inside is strong and uniform.
- Field outside is very weak (≈ 0).
- Solenoid behaves like a bar magnet.

Right Hand Grip Rule

Curl the fingers in the direction of current in the turns, thumb gives the direction of $\vec{B}$ (inside solenoid).



SUMMARY TABLE

Situation	Magnitude of $\vec{B}$	Direction
Long straight wire	$\frac{\mu_0 I}{2\pi r}$	Right hand thumb rule
Circular loop (centre)	$\frac{\mu_0 I}{2R}$	Right hand curl rule
Loop (on axis)	$\frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$	Along axis
Solenoid (inside)	$\mu_0 n I$	Right hand grip rule

Remember: Moving charge $\rightarrow$ Current $\rightarrow$ Magnetic field (Oersted's experiment).
Magnetism is a beautiful example of how electricity and magnetism are interconnected!

7 AMPÈRE'S CIRCUITAL LAW

The line integral of magnetic field $\vec{B}$ over any closed path (loop or surface) is equal to μ_0 times the net current enclosed by the path.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

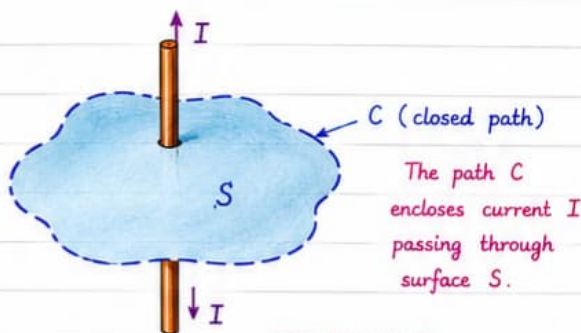
Where,

$\vec{B}$ = magnetic field

$d\vec{l}$ = infinitesimal length element along the path

I_{enclosed} = net current enclosed

μ_0 = permeability of free space



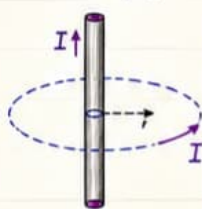
The path C encloses current I passing through surface S.

Special Cases

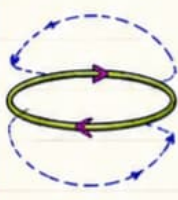
- ★ If no current is enclosed, $\oint \vec{B} \cdot d\vec{l} = 0$
- ★ If path is an open curve, $\oint \vec{B} \cdot d\vec{l}$ has no meaning.

Applications

Long straight current carrying wire



Circular current loop



Solenoid



Toroid



Key Points

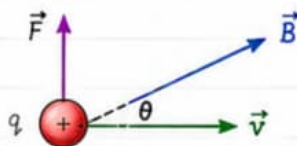
- ★ Works for steady (constant) currents only.
- ★ Basis of calculation of $\vec{B}$ in many symmetric situations.
- ★ Very powerful tool in magnetostatics.

8 FORCE ON A MOVING CHARGE IN A MAGNETIC FIELD

A charge q moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$ experiences a magnetic force $\vec{F}$ given by Lorentz force law.

$$\vec{F} = q(\vec{v} \times \vec{B})$$

$$|\vec{F}| = |q|vB \sin \theta$$



$\vec{v}$ = velocity of charge

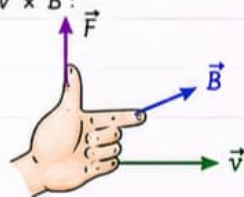
$\vec{B}$ = magnetic field

θ = angle between $\vec{v}$ and $\vec{B}$

(Right hand rule)

Direction

Use right hand rule for $\vec{v} \times \vec{B}$:



Special Cases

- ★ If $\vec{v} \parallel \vec{B}$ ($\theta = 0^\circ$ or 180°) $\Rightarrow F = 0$ (no force)
- ★ If $\vec{v} \perp \vec{B}$ ($\theta = 90^\circ$) $\Rightarrow F = |q|vB$ (maximum)
- ★ If charge is negative, direction of $\vec{F}$ is opposite.

Important

Magnetic force does no work on a charge, because $\vec{F} \perp \vec{v} \Rightarrow \vec{F} \cdot \vec{v} = 0$

It only changes the direction, not the speed.

(Kinetic energy remains constant)

Example

An electron moves with velocity $\vec{v}$ perpendicular to $\vec{B}$. Find the magnitude and direction of force.

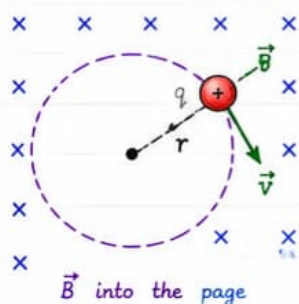
Solution:

$F = evB$ (since $\theta = 90^\circ$)

Direction: Use right hand rule (for negative charge, opposite).

9 MOTION OF A CHARGE IN A UNIFORM MAGNETIC FIELD

When a charged particle enters a uniform magnetic field perpendicular to $\vec{B}$, it moves in a circular path.



Derivation

Magnetic force provides centripetal force.

$$qvB = \frac{mv^2}{r}$$

$$\Rightarrow r = \frac{mv}{|q|B}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{|q|B}$$

T = time period

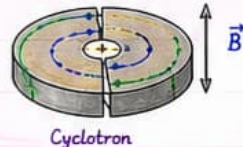
r = radius of circular path

Summary Table

Quantity	Formula
Radius (r)	$r = \frac{mv}{ q B}$
Time period (T)	$T = \frac{2\pi m}{ q B}$
Frequency (f)	$f = \frac{1}{T} = \frac{ q B}{2\pi m}$
Speed (v)	$v = \frac{ q Br}{m}$

Key Points

- ★ Path is circular if $\vec{v} \perp \vec{B}$.
- ★ Radius $\propto$ mass (m)
- ★ Radius $\propto$ velocity (v)
- ★ Radius $\propto \frac{1}{B}$
- ★ Independent of charge sign (only direction changes).
- ★ Used in cyclotrons, mass spectrographs,



Cyclotron



Remember

- Moving charge $\rightarrow$ Magnetic field
- Changing magnetic field $\rightarrow$ Electric field (Chapter 6)
- Current $\rightarrow$ Magnetic field



10 EFFECT OF MAGNETIC FIELD ON A CURRENT-CARRYING CONDUCTOR

A current-carrying conductor placed in a magnetic field experiences a force. This is the principle behind electric motors.

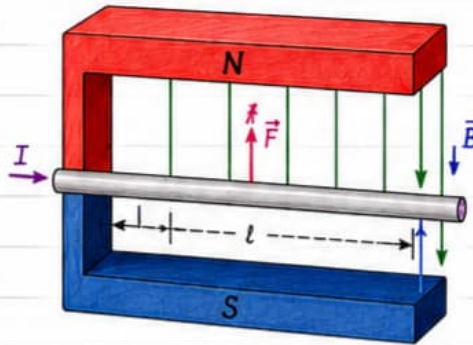
$$\vec{F} = B I l \sin \theta$$

B = magnetic field

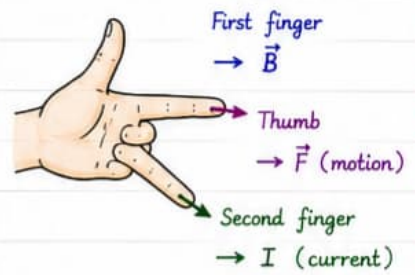
I = current

l = length of conductor in magnetic field

θ = angle between I and $\vec{B}$



Direction (Fleming's Left Hand Rule)



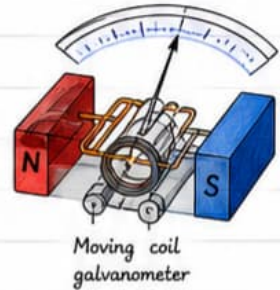
Special Cases

★ If $\theta = 0^\circ$ or 180° ($I \parallel \vec{B}$) $\Rightarrow F = 0$ (no force)

★ If $\theta = 90^\circ$ ($I \perp \vec{B}$) $\Rightarrow F = BIl$ (maximum)

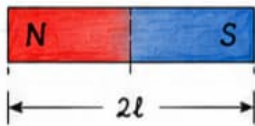
Applications

- Electric motors
- Moving coil galvanometer
- Loudspeakers
- Magnetic flow meters



11 MAGNETIC DIPOLE AND ITS FIELD

A bar magnet has two poles (N and S) separated by a distance $2l$. It behaves like a magnetic dipole.



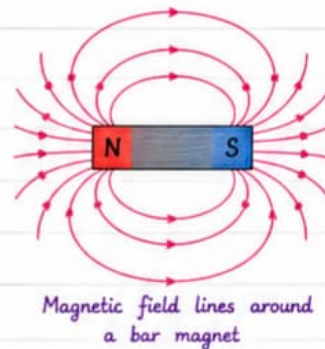
Magnetic dipole moment

$$\vec{m} = IA\hat{n}$$

For bar magnet,

$$m = M \times 2l$$

Direction: from S to N

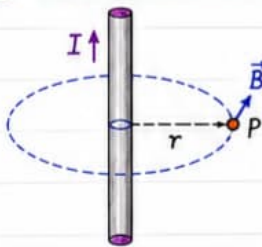


Magnetic field lines around a bar magnet

Key Points

- ★ Field lines emerge from N pole and enter S pole outside the magnet.
- ★ Inside the magnet, field lines go from S to N.
- ★ No two field lines intersect.
- ★ Density of field lines shows strength of field.

12 MAGNETIC FIELD DUE TO A STRAIGHT CURRENT-CARRYING CONDUCTOR (RECAP)



$$B = \frac{\mu_0 I}{2\pi r}$$

(direction: right hand thumb rule)

Note

- $B \propto I$ (directly)
- $B \propto \frac{1}{r}$ (inversely)
- Field lines are concentric circles around the wire.

Units

- $B \rightarrow$ Tesla (T) = Weber/m²
 $I \rightarrow$ Ampere (A)
 $l \rightarrow$ metre (m)
 $F \rightarrow$ Newton (N)
 $m \rightarrow$ A·m² (magnetic dipole moment)

SUMMARY TABLE

Situation	Expression	Direction	Rule
Magnetic field due to moving charge	$\vec{B} = \frac{\mu_0}{4\pi} \frac{q(\vec{v} \times \hat{r})}{r^2}$	$\vec{v} \times \hat{r}$	Right hand rule
Magnetic field due to current element	$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$	$d\vec{l} \times \hat{r}$	Right hand rule
Biot - Savart law	$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$	$d\vec{l} \times \hat{r}$	Right hand rule
Force on moving charge	$\vec{F} = q(\vec{v} \times \vec{B})$	$\vec{v} \times \vec{B}$	Right hand rule
Force on current-carrying conductor	$\vec{F} = BIl \sin \theta$	Fleming's left hand rule	Fleming's left hand rule
Magnetic field (straight wire)	$B = \frac{\mu_0 I}{2\pi r}$	Right hand thumb rule	Right hand thumb rule
Magnetic dipole moment	$\vec{m} = IA\hat{n}$ (bar magnet: $m = M \times 2l$)	S $\rightarrow$ N	—

Quick Revision

- ✓ Moving charges produce magnetic field.
- ✓ Current produces magnetic field.
- ✓ Changing $\vec{B}$ produces electric field (next chapter).
- ✓ Direction is everything - remember the rules!

Memory Booster

"Current creates B ,
 Changing B creates E -
 That is the cycle
 of Electromagnetism!"



Remember

Magnetism is not a force, it is a field. Visualize the field, apply the right rule, and the answer is yours! 😊

All the Best

